

I'm trying to build my computer vision skills to prepare for a job that applies computer vision to healthcare industry. The job also asks for experience with the NVIDIA tools such as NeMo and Nemotron. Is there a sample project I can build that would help build my skills and familiarity in this area? The skin cancer dataset at <https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000> seems interesting to me.

Yes. HAM10000 is an excellent dataset for this goal, particularly because you can turn it into a project that goes well beyond "train a CNN to classify skin lesions." The strongest portfolio project would combine computer vision, medical-AI evaluation, explainability, and a vision-language model, while giving you a reason to work with NVIDIA's current NeMo/Nemotron ecosystem. HAM10000 contains 10,015 dermoscopic images covering seven diagnostic categories, with ground truth established through pathology, follow-up, expert consensus, or confocal microscopy. Nature A project I'd recommend "Multimodal Skin Lesion Analysis with Computer Vision and NVIDIA NeMo" Build an end-to-end prototype that takes a dermoscopic image and produces something like: Predicted category: Melanoma Confidence: 87% Visual evidence: Irregular pigmentation and asymmetric lesion structure Model explanation: ... Important caveat: This is a research/educational model and is not a clinical diagnosis.

I'd structure it as five progressively more sophisticated stages.

1. Build a conventional computer-vision baseline

Start with HAM10000 and build a proper image-classification pipeline. For example: Input dermatoscopy image → preprocessing → neural network → 7-class prediction Try progressively more sophisticated models:

1. Simple CNN
2. ResNet/EfficientNet
3. Vision Transformer
4. Transfer learning from a pretrained model

Don't just report accuracy. Because HAM10000 is highly imbalanced, I'd evaluate:

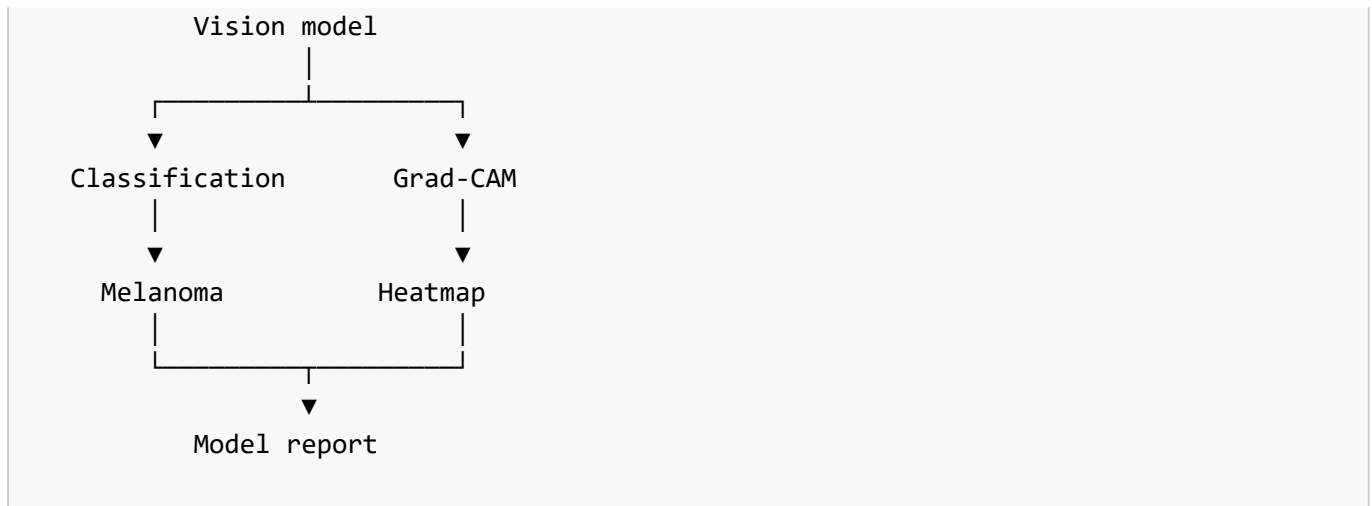
- balanced accuracy
- macro F1
- per-class precision/recall
- confusion matrix
- ROC-AUC where appropriate
- sensitivity for melanoma
- specificity for melanoma This alone gives you a legitimate computer-vision project. But I'd deliberately not stop here.

2. Add medical-imaging-specific explainability

This is where the project starts becoming much more interesting for a healthcare-CV job. Add Grad-CAM or another saliency technique and visualize: Original image → model prediction → heatmap showing where the model was looking For example:

```
graph TD; A[HAM10000 image] --> B[▼];
```

HAM10000 image
↓
▼



Then investigate an important healthcare-AI question: Is the model looking at the lesion, or is it exploiting artifacts/background features?

That is a much more impressive question to investigate than simply maximizing accuracy. You could deliberately test:

- images with borders/cropping
- different image backgrounds
- image artifacts
- hair
- lesion size
- acquisition source HAM10000 is particularly suitable for this because its images came from different populations, modalities, and two clinical sites. Nature

3. Add a Vision-Language Model

This is the part I'd use to build your NeMo/Nemotron familiarity. Instead of having the system output only: melanoma: 0.87

have it produce a structured interpretation: Diagnosis prediction: Melanoma

Confidence: 0.87

Visual observations:

- Asymmetric lesion
- Irregular pigmentation
- Irregular border

Potentially concerning characteristics:

- Asymmetry
- Color variation

Important limitations: The model is trained on dermoscopic images and should not be used as a clinical diagnostic system. You can use a VLM to generate the natural-language portion while keeping the conventional CV model responsible for the actual classification. That's particularly relevant to NVIDIA's current tooling. NeMo Framework now encompasses multimodal and computer-vision workloads, in addition to

language models. NVIDIA Docs And current NeMo AutoModel supports VLM training/fine-tuning, including image-text-to-text models. NVIDIA Docs.

4. Create your own multimodal training dataset

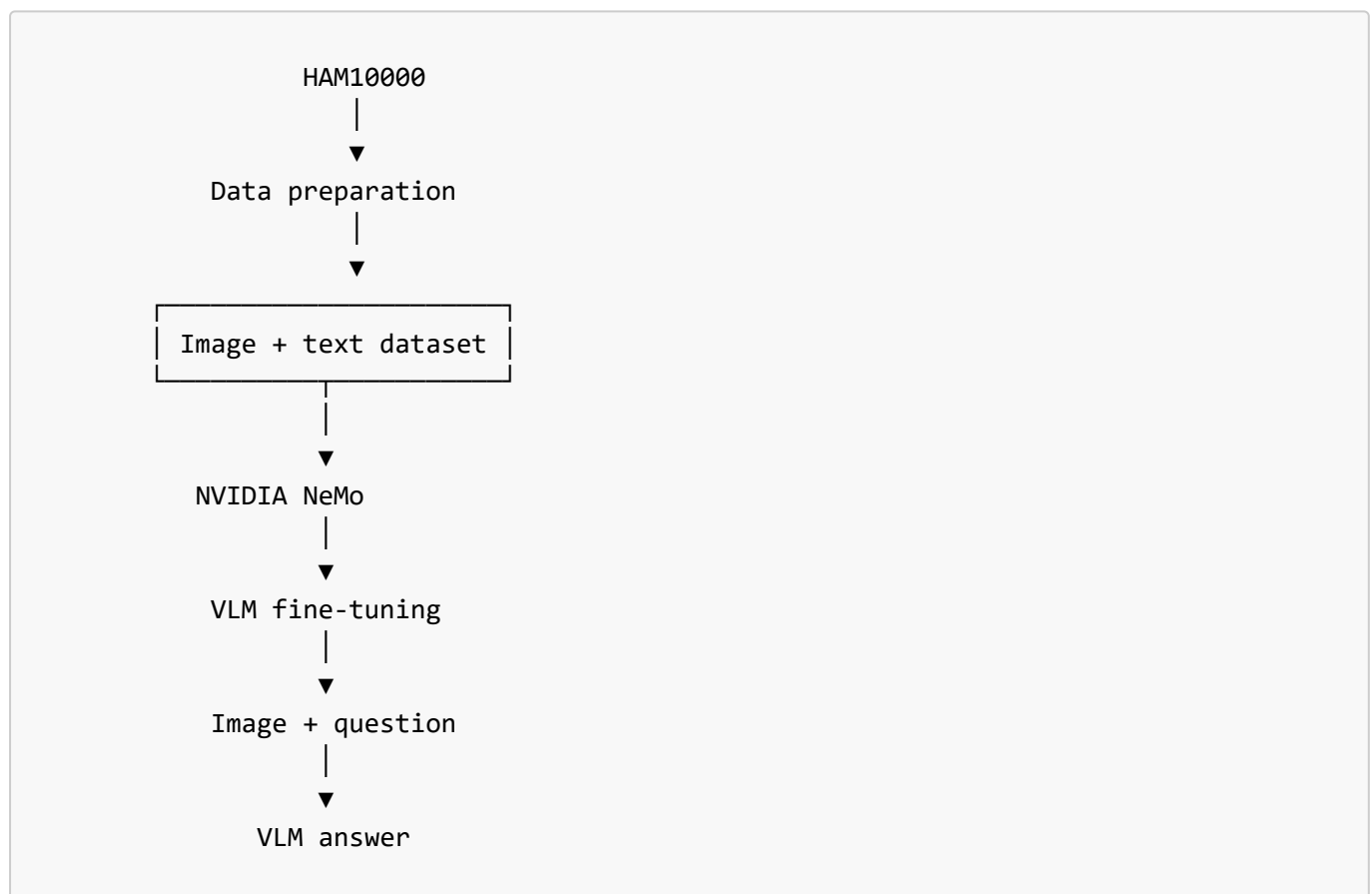
This could be the most educational part of the project. HAM10000 doesn't naturally give you the sort of question/answer data needed to train a VLM. So construct a derived dataset.

For example: { "image": "ISIC_0027419.jpg", "question": "What type of lesion is shown?", "answer": "Melanocytic nevus" } Then expand it: { "image": "ISIC_0027419.jpg", "question": "What visual characteristics are present?", "answer": "The lesion is relatively symmetric with fairly uniform pigmentation." } And: { "image": "ISIC_0027419.jpg", "question": "What diagnostic category does this lesion belong to?", "answer": "Melanocytic nevus." } You could generate multiple question types from the HAM10000 metadata and your CV model. This would give you experience with an important real-world problem: How do you turn an existing computer-vision dataset into a multimodal dataset suitable for foundation-model fine-tuning?

That's directly relevant to current NeMo workflows. NVIDIA's documentation specifically describes integrating custom image + text datasets using preprocessing and YAML configuration. NVIDIA Docs

5. Fine-tune a VLM with NeMo

Then bring NVIDIA explicitly into the project. The workflow becomes:



NeMo AutoModel currently supports PEFT/LoRA fine-tuning for supported VLMs, which makes this considerably more approachable than trying to train a large model from scratch. NVIDIA Docs And NVIDIA's current documentation actually has an end-to-end example of fine-tuning a VLM on a medical VQA dataset using NeMo AutoModel. NVIDIA Docs That's a particularly useful reference for your project because the

NVIDIA example uses MedPix-VQA, a medical image/question/answer dataset. Where Nemotron fits I would not force Nemotron into the first version just because the job description mentions it. Instead, make it the next stage. For example:

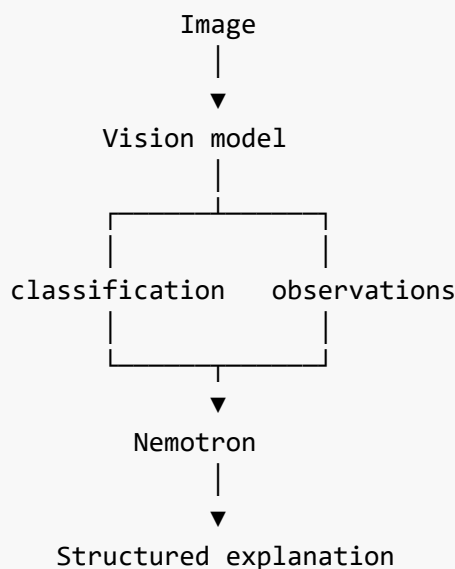
Stage A — CV model

Image → ResNet/ViT → diagnosis

Stage B — VLM

Image + question → VLM → answer

Stage C — Nemotron-powered medical assistant



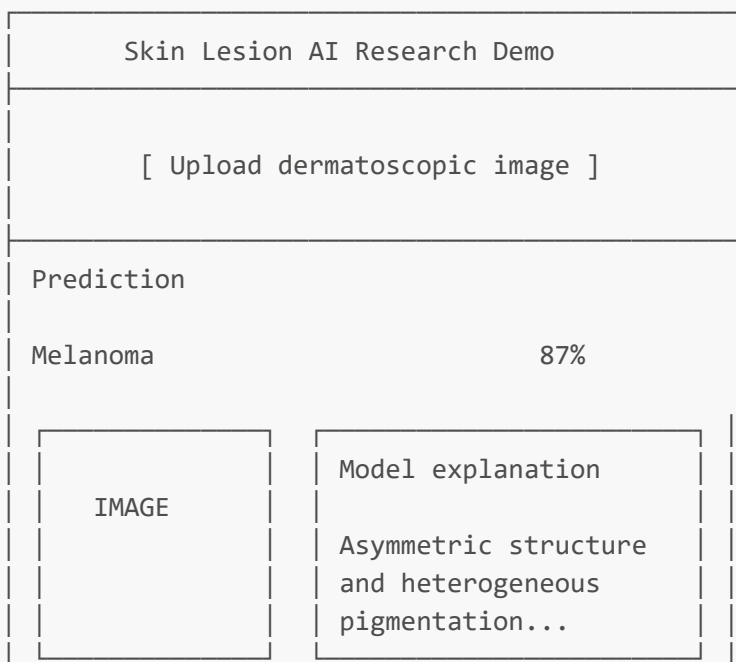
The language model could take the structured outputs from your CV system and turn them into a coherent report. That gives you an opportunity to demonstrate that you understand an important distinction: A language model isn't necessarily the right model for the image classification task. You can use specialized CV models for visual perception and an LLM/VLM for reasoning, interaction, and report generation. That's much closer to how I'd expect a sophisticated healthcare AI system to be architected. One particularly interesting experiment I'd make this a central research question: Can a VLM identify clinically relevant visual features without being explicitly told what to look for? For example, give it an image and ask: "Describe the visual characteristics of this lesion."

Then compare its response against:

- the actual HAM10000 diagnosis
- metadata
- your CV model's prediction
- Grad-CAM
- perhaps dermatologist-created descriptions from another dataset You could investigate whether the VLM's explanations correlate with the features that actually drive the classifier. That's a genuinely interesting CV/AI experiment rather than just a tutorial. Your final portfolio project could look like this I'd aim for a GitHub repository something like:

```
ham10000-nvidia-medical-ai/
├── data/
│   └── README.md
├── notebooks/
│   ├── 01_dataset_exploration.ipynb
│   ├── 02_baseline_cnn.ipynb
│   ├── 03_transfer_learning.ipynb
│   ├── 04_model_evaluation.ipynb
│   └── 05_gradcam_analysis.ipynb
├── src/
│   ├── data/
│   ├── models/
│   ├── evaluation/
│   ├── explainability/
│   └── vlm/
├── nemo/
│   ├── dataset.yaml
│   ├── model.yaml
│   └── train.sh
├── app/
│   └── demo.py
├── README.md
└── results/
```

And ultimately have a small web application:



[Show Grad-CAM] [Ask VLM] [Compare Models]

That would give you something demonstrable in an interview, rather than simply saying "I trained a model on HAM10000." How I'd sequence your learning Given your goal, I would not start with NeMo. I'd deliberately build the project in this order:

Phase	Skill you develop	Technology
1	Medical image preprocessing	Python/OpenCV/PyTorch
2	CNN image classification	PyTorch
3	Transfer learning	ResNet/ViT
4	Medical ML evaluation	sklearn/PyTorch
5	Explainable CV	Grad-CAM
6	Vision-language models	Transformers
7	Multimodal dataset construction	JSON/YAML/Python
8	VLM fine-tuning	NeMo AutoModel
9	NVIDIA model ecosystem	Nemotron
10	Deployment	NIM/Triton/etc.

This progression is useful because each stage teaches you something fundamental rather than hiding the computer-vision concepts behind a foundation model. NVIDIA's current NeMo ecosystem also includes tools for deploying models and NIM provides managed/self-hosted inference interfaces for VLMs, so deployment can eventually become another part of the project. NVIDIA Docs One important caveat I'd describe the project as medical-AI research/education, not as a skin-cancer diagnostic system. HAM10000 is a valuable benchmark, but its labels and collection process have limitations, and performance on this dataset wouldn't establish clinical validity. The dataset authors themselves describe it as a training set for academic machine learning purposes. Nature I think this could be an unusually good project for the job you're targeting because it lets you demonstrate three different capabilities in one coherent application: computer vision → medical AI → NVIDIA generative/multimodal AI. If you'd like, I can take this one step further and lay out a concrete 6–8 week project plan, including the specific PyTorch models, NVIDIA/NeMo/Nemotron models I'd use, datasets, milestones, and what you'd want to have in the GitHub repo at the end.